

The Economics of Bicycles for the Mind

Cognitive tools, human judgment, and inequality

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Agrawal, Gans & Goldfarb (2025) · NBER WP 34034

github.com/gsaco/ai-02-agrawal

The paper and the agent's problem

Question and single mechanism

Question. When a cognitive tool makes implementation easier, how do effort, productivity, returns to human abilities, and inequality change?

Mechanism. The tool is a **substitute for implementation effort** at the margin; implementation skill s , payoff judgment α , and opportunity judgment $\gamma(t)$ determine who can exploit it.

Agent's problem, conditional on finding an opportunity

Choose effort e_t to maximize expected improvement value net of cost:

$$e_t^*(\theta) \in \arg \max_{e_t \geq 0} \{p(se_t; \theta)\alpha\Delta - c(e_t; \theta)\}.$$

Interior FOC: $p'(se_t^*; \theta)s\alpha\Delta = c'(e_t^*; \theta)$.

Choice and constraint: $e_t \geq 0$.

Technology: $p \in [0, 1]$ is increasing and weakly concave.

Cost: c is increasing and weakly convex.

Tool: raises p and/or lowers c , while lowering p'/c' .

Proposition 3: the conditional variance result

Specialization and conditions

$p(se; \theta) = \sqrt{se + \theta}$, $c(e) = e$; $\Gamma = \gamma_0/(1 - \delta\gamma)$ and $\delta\gamma < 1$.

$\alpha, \gamma_0, \gamma, s$ are independent, have positive support, and satisfy $\mu_i > 3\sigma_i$.

Common interior range: $\theta < \alpha^2 \Delta^2 s^2 / 4$ for every worker under comparison.

Corrected exact statement

$$V(\theta) = \Gamma \left(\frac{\alpha^2 \Delta^2 s}{4} + \frac{\theta}{s} \right).$$

$$\frac{d\mathbb{E}[V]}{d\theta} = \mathbb{E}[\Gamma] \mathbb{E}[1/s] > 0.$$

The U-shape concerns cross-sectional $\text{Var}[V(\theta)]$ as continuous tool quality θ changes.

If

$$\frac{\mathbb{E}[\Gamma^2]}{\mathbb{E}[\Gamma]^2} < \mu_s \mathbb{E}[1/s], \quad \text{Var}(\Gamma/s) > 0,$$

then $\text{Var}[V(\theta)]$ is strictly convex, falls for $\theta < \theta^*$, and rises for $\theta > \theta^*$, where $\theta^* = -a_0/[2 \text{Var}(\Gamma/s)] > 0$.

Endpoint correction and intuition

Equation (32)'s positive slope at $\theta = 1$ also requires $\theta^* < 1$. **Intuition:** inverse skill bias equalizes first; heterogeneous opportunity judgment can dominate later.

What I did: interpreting the U-shape

1. The U-shape is in what, and with respect to what?

Object: $\text{Var}_i[V_i(\theta)]$, the cross-sectional variance across workers of total continuation value. **Axis:** tool quality θ .
Domain: the common-interior range—not time-series or simulation-draw variance.

2. Conditional total variance; monotone benefit variance

Continuation-value variance.

$$\frac{d \text{Var}_i[V_i(\theta)]}{d\theta} = a_0 + 2\theta \text{Var}_i(\Gamma_i/s_i).$$

Condition (30) gives $a_0 < 0$; heterogeneity in Γ_i/s_i gives positive curvature. An upturn by $\theta = 1$ additionally requires $\theta^* < 1$.

Variance of the individual benefit.

$$B_i(\theta) = V_i(\theta) - V_i(0) = \theta \Gamma_i/s_i,$$

$$\text{Var}_i[B_i(\theta)] = \theta^2 \text{Var}_i(\Gamma_i/s_i).$$

This variance rises monotonically for $\theta \geq 0$; it is not U-shaped.

3. What I explored, what came out, and the dead ends

Assumptions. I questioned the endpoint claim, continuous versus binary θ , common interior, a probability cap, and a general opportunity path; I kept independence and positive support.

Dead ends. Beyond the common interior, zero-effort corners make the formula piecewise; a probability cap or a general opportunity path changes the model. I therefore do not claim the same global U-shape.

Evidence. `sim.py` used SymPy and exact moments—not Monte Carlo. Condition (30) holds, yet $\theta^* = 1.7007$, the slope at one is -0.0051 , and the interior limit is 2.25. Lean confirms that the missing endpoint condition is $\theta^* < 1$.

Hand check: where I questioned the AI

expected net benefit: $M(e, \theta) = \underbrace{p(se, \theta)\alpha\Delta}_{\text{expected benefit}} - \underbrace{c(e, \theta)}_{\text{cost}}$

the worker chooses optimal effort: $e^*(\theta) = \arg \max_{e \geq 0} M(e, \theta)$

$\frac{\partial M}{\partial e} = p'(se, \theta)\alpha\Delta - c'(e, \theta) = 0 \rightarrow p'(se^*, \theta)\alpha\Delta = c'(e^*, \theta)$

$M^*(\alpha, \theta) = M(e^*(\alpha, \theta), \alpha, \theta) \rightarrow \frac{dM^*}{d\alpha} = \underbrace{\frac{\partial M}{\partial \alpha}}_{\text{direct effect}} + \underbrace{\frac{\partial M}{\partial e} \cdot \frac{\partial e^*}{\partial \alpha}}_{\text{indirect effect}}$

* $\frac{\partial M}{\partial \alpha} = p(se^*, \theta)\Delta \rightarrow \frac{dM^*}{d\alpha} = p(se^*, \theta)\Delta$

$\frac{\partial M}{\partial e} = p'(se^*, \theta)\alpha\Delta - c'(e^*, \theta) = 0$ at the optimum

10 This is the envelope theorem result used by authors in the the proof of proposition 2.

My handwritten check: direct effect + indirect effort response.

What the AI answered

$$\frac{\partial [V_0(1) - V_0(0)]}{\partial \alpha} = \Gamma \Delta (p_1^* - p_0^*)$$

where $p_\theta^* = p(se^*(\theta); \theta)$.

What I verified by hand

For $M = p(se; \theta)\alpha\Delta - c(e; \theta)$ and an interior optimum, the FOC kills the indirect effort response:

$$\begin{aligned} \frac{dM^*}{d\alpha} &= \frac{\partial M}{\partial \alpha} + \frac{\partial M}{\partial e} \frac{\partial e^*}{\partial \alpha} \\ &= p(se^*; \theta)\Delta. \end{aligned}$$

Verdict: correct under interior differentiability

Correct. At $\theta = 1$ and $\theta = 0$, subtract and multiply by Γ : this is exactly the AI identity.

But only this. It gives no sign for $p_1^* - p_0^*$ and therefore no global complementarity, adoption, or inequality result. At $e^* = 0$ the photographed FOC proof does not apply.